

Statistical analysis on depression, anxiety and stress scales (DASS21)

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Abstract

This study investigates psychological distress among Portuguese university students using the DASS-21 scale, focusing on depression, anxiety, and stress. A one-way ANOVA was first conducted to examine differences in distress levels across age groups, revealing a significant effect for stress but not for depression or anxiety. An exploratory factor analysis (EFA) confirmed the three-factor structure of the DASS-21 and demonstrated strong internal consistency, despite a few weaker items. Finally, a clustering analysis identified two primary psychological profiles, students with low and high levels of distress, highlighting the presence of distinct mental health patterns within the population.

Keywords: DASS-21, depression, anxiety, stress, One-way Anova, Exploratory Factor analysis, Clustering

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Introduction

The Depression, Anxiety, and Stress Scales (DASS) is a set of three self-report instruments designed to measure the negative emotional states of depression, anxiety, and stress (Parkitny & McAuley, 2010). Developed to aid in the definition, assessment, and differentiation of these emotional conditions, the DASS has demonstrated strong psychometric properties, including high internal consistency and the ability to distinguish between emotional states in a variety of clinical and research settings (Oei et al., 2013). These qualities make the DASS suitable for use by both researchers and clinical practitioners.

The original DASS consists of 42 items, but a shorter version, DASS-21, has also been developed. This abbreviated version includes 21 items, with 7 questions corresponding to each of the three scales (Oei et al., 2013). In the present work, we use the DASS-21 to assess emotional states. Each item is rated on a four-point Likert scale, ranging from 0 ("Did not apply to me at all") to 3 ("Applied to me very much or most of the time"), based on how the respondent has felt over the past week. Scores for each scale are obtained by summing the responses to the corresponding seven items, either for individual analysis or to compute an overall measure of emotional distress.

High scores on each scale are associated with the following characteristics (Psychology Foundation of Australia, 2025):

- **Depression scale:** self-disparaging, dispirited, gloomy, blue, convinced that life has no meaning or value, pessimistic about future, unable to experience enjoyment or satisfaction, unable to become interested or involved, slow, lacking in initiative
- **Anxiety scale:** apprehensive, panicky, trembly, shaky, aware of dryness of the mouth, breathing difficulties, pounding of the heart, sweatiness of the palms, worried about performance and possible loss of control
- **Stress scale:** over-aroused, tense, unable to relax, touchy, easily upset, irritable, easily startled, nervy, jumpy, fidgety, intolerant of interruption or delay

The internal consistency of the DASS-21 subscales has been reported as high, with Cronbach's alpha values typically ranging from 0.96 to 0.97 for depression, 0.84 to 0.92 for anxiety, and 0.90 to 0.95 for stress (Parkitny & McAuley, 2010).

In this study, we apply a range of statistical methods—including Analysis of Variance (ANOVA) and factorial analysis—to examine how DASS-21 scores vary across different groups or conditions (more details about the population on the next section). These methods allow us to explore the relationships between emotional states and various independent variables, as well as interactions among them, providing insights into patterns of psychological well-being across the sample.

Demographic overview

The DASS-21 questionnaire was administered to university students in Portugal. The dataset comprises 226 valid responses collected from a diverse group of students. While the vast majority of participants are Portuguese nationals, only six respondents hold other nationalities, though all originate from countries where Portuguese is the official language.

In terms of age distribution, most participants are between 19 and 24 years old, reflecting the typical university age population. However, the dataset also includes responses from older students, with some participants over the age of 50, highlighting the diversity within the Portuguese higher education system. The full age distribution is illustrated in Figure 1.

Regarding gender, the majority of respondents identify as female, followed by male participants, with a small number identifying as non binary. This distribution is shown in Figure 2.

The educational background of respondents is another important demographic factor. As presented in Figure 3, the majority, approximately 140 out of 226 participants, have completed secondary education. The remaining participants have pursued higher education, including bachelor's degrees, master's degrees, and even PhDs.

We also examined family income, as illustrated in the boxplot in Figure 4. The distribution suggests that most respondents come from middle level income families, though there are a few outliers at both the lower and upper ends of the income spectrum.

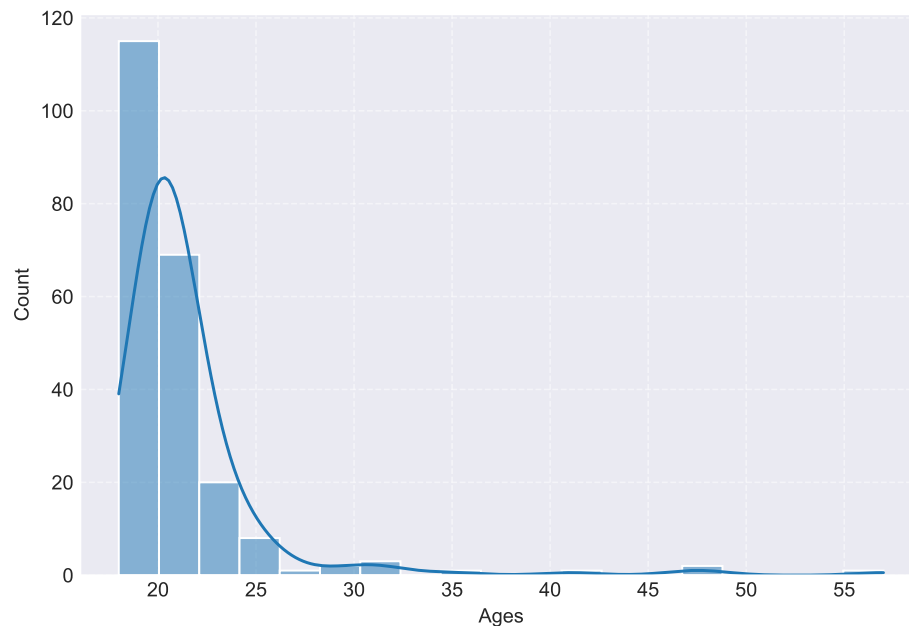


Figure 1

Age distribution of the population under study.

When comparing these findings with national level data provided by Eurostat (Eurostat, 2025), distributed in a conveniently representation by Pordata (Pordata, 2025), we find that the dataset presents a reasonably representative sample of the Portuguese university student population in terms of age, gender, education level, and socioeconomic status.

How to read this work

This project was conducted within the scope of the Research Methods II course, and its primary audience includes professors and students involved in this class. However, readers with a general interest in psychological assessment or statistical analysis may also find the work informative and relevant.

The remainder of this document is organized into four main sections:

- **Analysis of Variance - *One-Way ANOVA*:** This section explores whether there are significant differences in depression, anxiety, and stress levels across different age groups using one-way ANOVA.

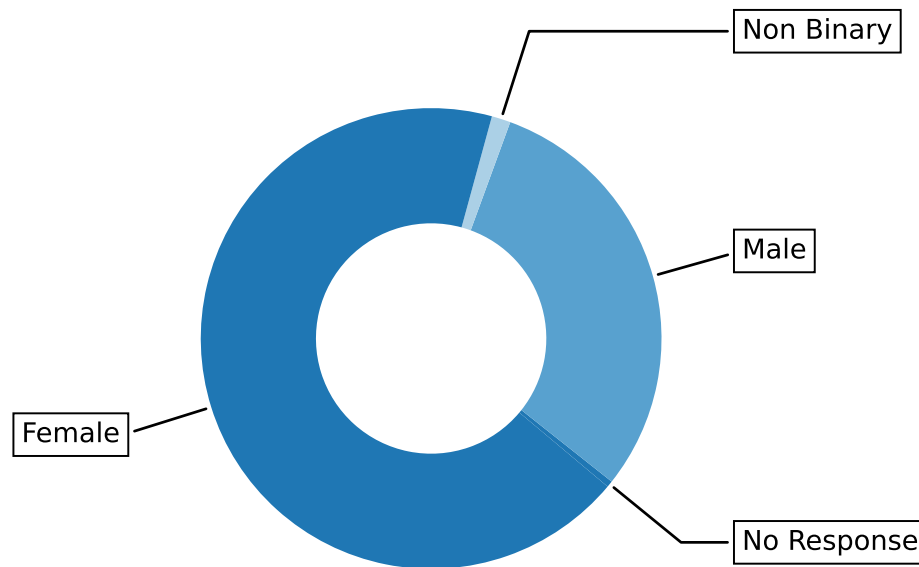


Figure 2

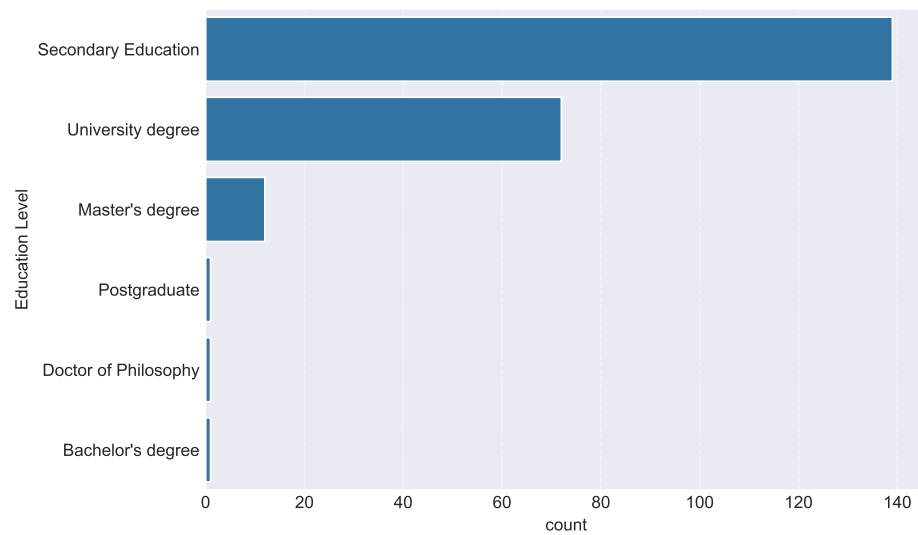
Gender distribution of the population under study.

- Exploratory Factor Analysis: EFA: Here, we verify the factor structure of the DASS-21 scale and evaluate the reliability of its components through EFA.
- Cluster analysis: In this section, we apply clustering techniques to identify distinct psychological profiles among university students based on their factor scores.
- Conclusion: This final section summarizes the main findings of the study and reflects on the implications of each analytical approach.

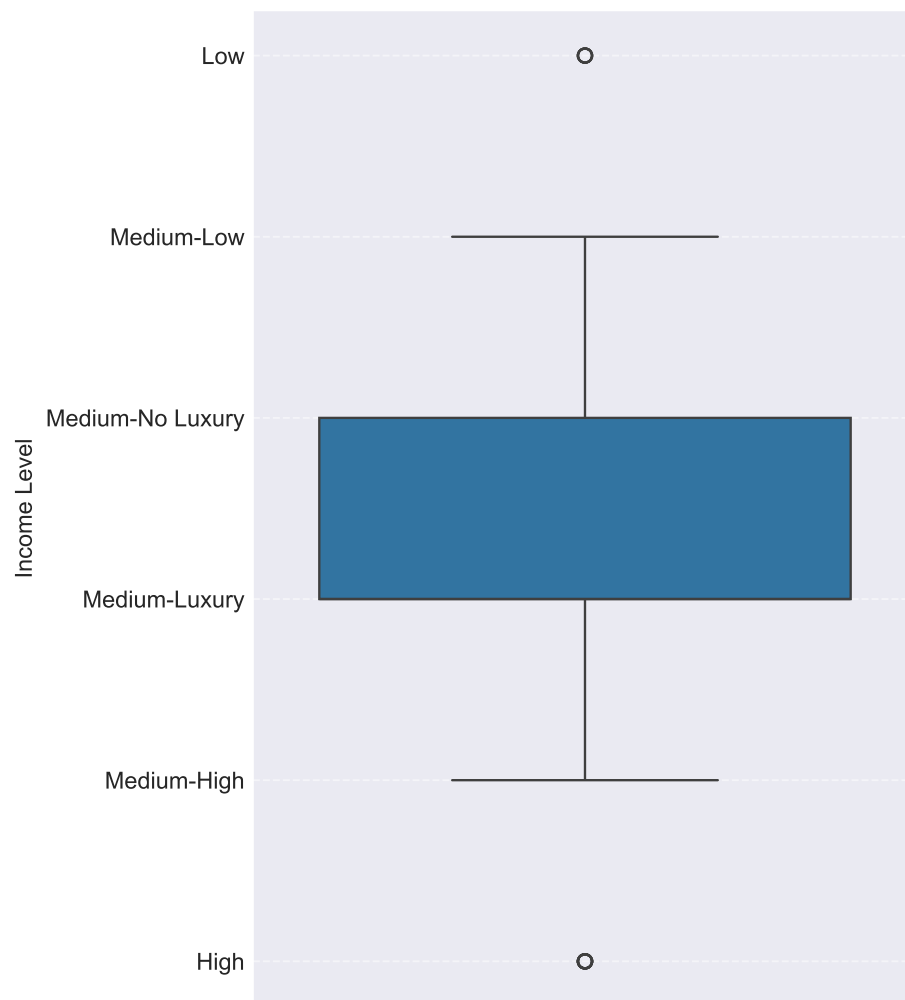
Tools used and replication packages

This document was written in $\text{\LaTeX}$ using the official APA 7th Edition style (Weiss, 2025).

All analyses presented in this work were conducted using Python in a Jupyter Notebook environment. Full replication instructions, including the list of required libraries and dependencies, are provided in Appendix A: Replication Package. Readers should be able to run the provided notebooks and reproduce all results shown in this document.

**Figure 3**

Education level of the population under study.

**Figure 4**

Family income level of the population under study.

Analysis of Variance - *One-Way ANOVA*

In this section, we examine how levels of depression, anxiety, and stress vary across different age groups of university students. As noted in the Introduction, the dataset contains an unbalanced number of responses for each age. To facilitate a fair comparison and ensure sufficient statistical power, we define the following age groups:

- **Group A (19 years old or younger):** 38 responses
- **Group B (between 20 and 21 years old):** 129 responses
- **Group C (22 years old or older):** 56 responses

Three respondents did not report their age. While it would be possible to conduct a t-test to compare those who provided their age against those who did not, in order to test for Missing Completely At Random (MCAR) (Costa et al., 2025b), we opt for a more pragmatic approach. Given that these represent only 1.3% of the total responses (3 out of 226), we exclude them from subsequent analysis to maintain simplicity without meaningfully affecting the results.

This grouping results in a sample size that is sufficiently large for conducting a one-way ANOVA (Keselman et al., 1998).

We set the following hypotheses for each of the DASS subscales (Depression, Anxiety, and Stress):

- **Null Hypothesis (H_0):** There is no significant difference in depression, anxiety or stress scores across age groups.
- **Alternative Hypothesis (H_1):** At least one age group has significantly different scores for depression, anxiety or stress.

This analysis helps us answer the research question:

“Do students from different age groups experience different levels of depression, anxiety and stress?”

The independent variable (IV) is age group, and the dependent variables (DVs) are the scores for depression, anxiety, and stress. Throughout the analysis, we use a significance level of $\alpha = 0.05$ (Houle et al., 2005).

Because the dataset includes some missing values, and follow-up with respondents, to understand why the values are blank, is not feasible, we use the Iterative Imputer from Scikit-learn¹ to impute missing values. This method estimates missing entries by modeling each variable as a function of the others in a round-robin fashion (Van Buuren & Groothuis-Oudshoorn, 2011).

Data validation

To ensure the validity of the one-way ANOVA, we verify the following assumptions:

- **Power Analysis:** To evaluate whether our sample size was sufficient to detect a statistically significant effect (Houle et al., 2005)
- **Independence of observations:** Each participant's response is independent of the others. This is assumed based on the survey's design, which ensured that each student submitted only one response.
- **Normality:** The distribution of scores in each age group should be approximately normal. We tested this using the Shapiro-Wilk test, as it is well-suited for small to moderate sample sizes (Costa et al., 2025b).
- **Homogeneity of variances:** The variance in each group should be approximately equal. We used Levene's test to check this assumption (Costa et al., 2025a).

Power Analysis

Given that the age groups in our dataset are unbalanced, we conducted a power analysis to assess the statistical power and the effect sizes our study is capable of detecting (Houle et al., 2005).

¹ <https://scikit-learn.org/stable/modules/generated/sklearn.impute.IterativeImputer.html>

Figure 5 presents a power curve analysis based on the total sample size. The plot displays power curves for different values of Cohen's f , a standardized measure of effect size for ANOVA. The red vertical dashed line represents our total sample size of 223 participants. This line intersects all curves corresponding to medium ($f = 0.25$) and large effect sizes ($f \geq 0.4$) at a power level exceeding 80%. This indicates that our sample size is sufficient to detect effects of at least medium magnitude with adequate statistical power.

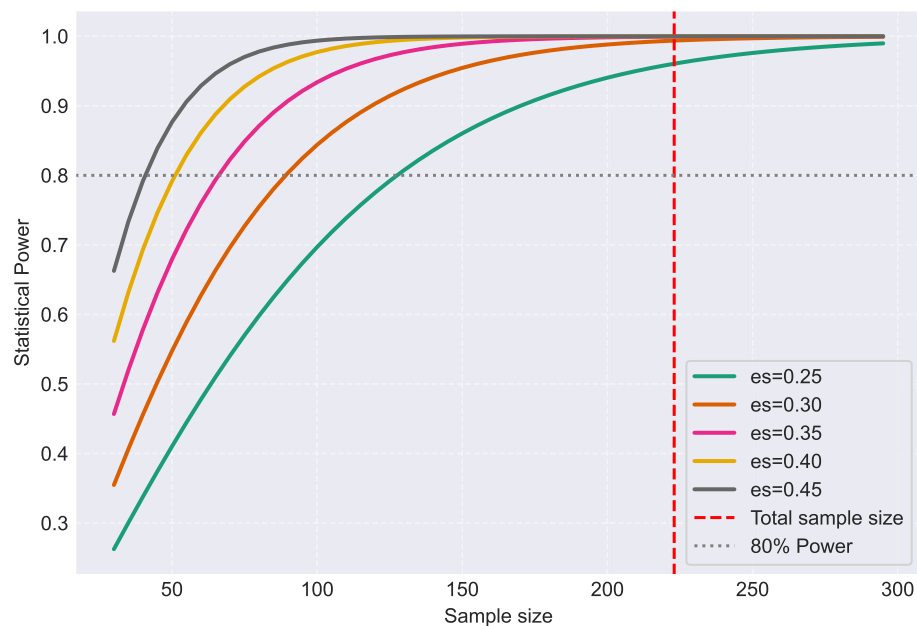


Figure 5

Power Curve for One-Way Anova (using total sample size)

To evaluate the adequacy of the individual group sizes, we extended our analysis to the groups. One limitation of the tool used to generate Figure 5 is that it assumes equal group sizes². Our dataset uses groups of unequal size.

To address this, we employed the harmonic mean of the group sizes as an approximation. This approach helps mitigate the overestimation that may occur if an arithmetic mean were used

² Details on the tools used for power analysis are available in Appendix A: Replication Package. The specific library and its documentation can be found at:

<https://www.statsmodels.org/dev/stats.html#power-and-sample-size-calculations>

in the presence of unequal groups. Figure 6 illustrates the power curve using the harmonic mean of the group sizes. For effect sizes greater than $f = 0.4$ (considered large), the statistical power exceeds 80%.

Additionally, when calculating the power using the actual group distribution, assuming a medium effect size of $f = 0.25$ and a significance level of $\alpha = 0.05$, the resulting statistical power is **92%**. This confirms that the analysis is sufficiently powered to detect medium-sized effects across the age groups.

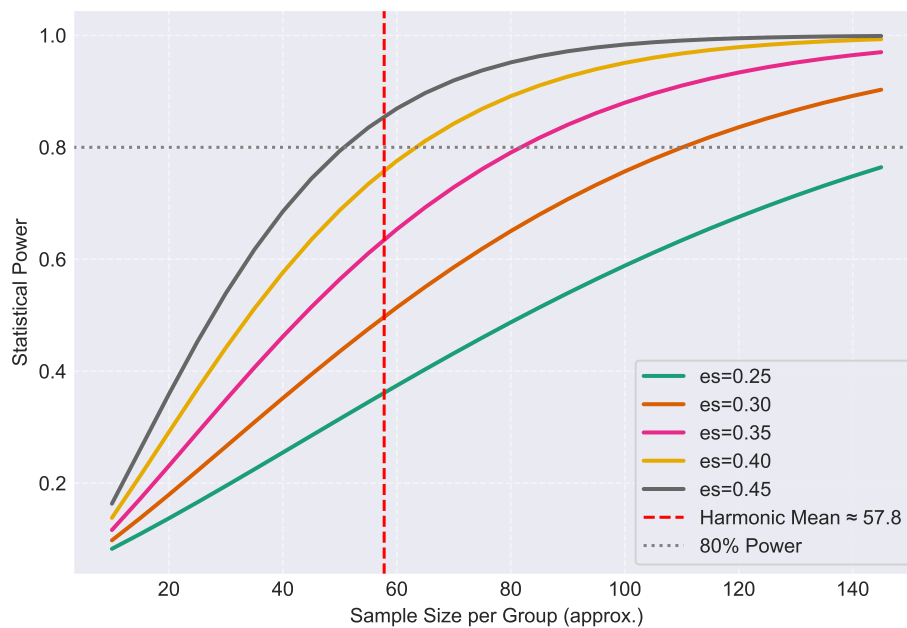


Figure 6

Power Curve for One-Way ANOVA (Using Harmonic Mean Approximation)

Normality and variance test

Figures 7, 8 and 9 display the score distributions for the three DASS-21 subscales across age groups. While some distributions approximate normality reasonably well, others show notable deviations. To investigate this further, we applied the Shapiro–Wilk test for normality.

Since only some groups clearly deviate from normality, we applied a Box–Cox transformation to stabilize variance and normalize the distributions³ (Box & Cox, 1964). Table 1

³ Transformation performed using the `scipy.stats.boxcox` function:

summarizes the Shapiro–Wilk test results after transformation.

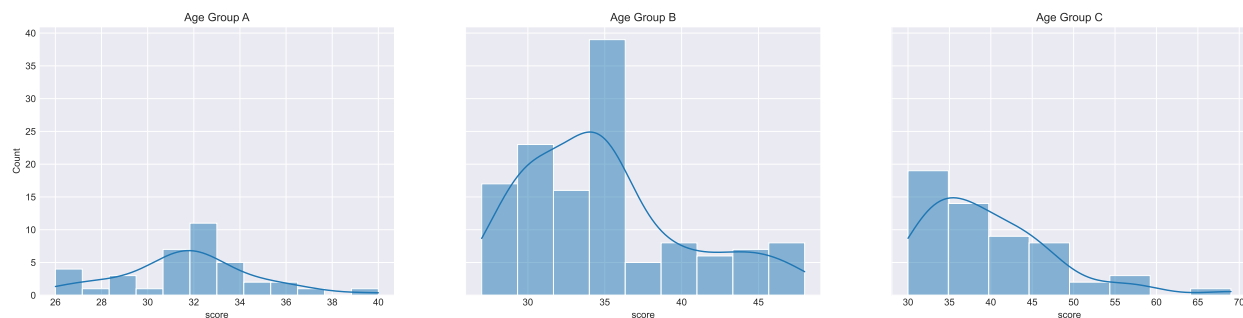


Figure 7

Depression scores distribution

Some groups present p-values below the 0.05 threshold, suggesting a statistically significant deviation from normality. Specifically, Group B and C for depression, Group B for anxiety, and Group A for stress. However, all W-statistics remain relatively high (close to 1), indicating that deviations from normality are relatively mild.

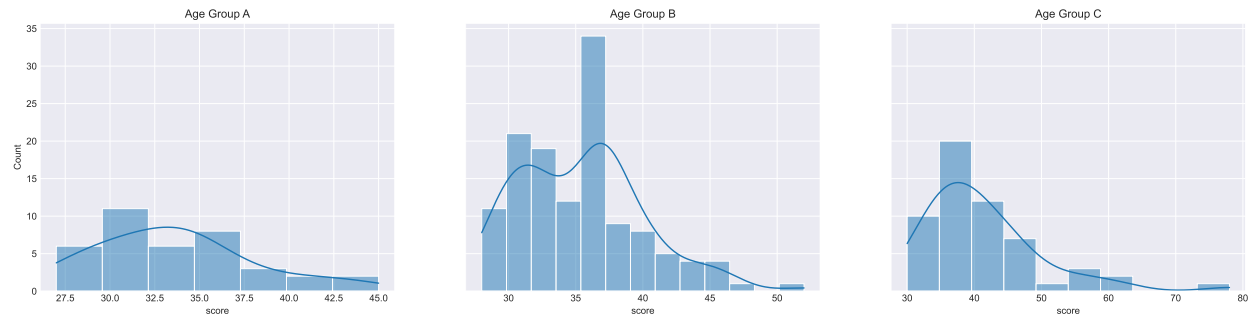
Because ANOVA is generally robust to small violations of normality, we proceeded to assess the assumption of homogeneity of variances using Levene's test. Results are shown in Table 2.

Levene's test indicates a significant violation of the homogeneity of variances assumption for the depression and anxiety scales ($p < 0.05$). In contrast, the stress scale does not show significant variance heterogeneity ($p = 0.0689$), suggesting that the assumption is met in that case.

In summary, the stress scale satisfies both key assumptions for conducting a one-way ANOVA. However, due to violations in normality and variance assumptions for the depression and anxiety scales, we will not proceed with ANOVA for those two. Instead, we revise our initial hypothesis and retain the null hypothesis of no significant differences across age groups for depression and anxiety.

Analysis

The results of the one-way ANOVA conducted on the stress scale are presented in Table 3.

**Figure 8***Anxiety scores distribution*

The p-value ($PR(>F)$) is significantly lower than the 0.05 threshold, indicating strong evidence against the null hypothesis. We therefore reject the null hypothesis and conclude that there are statistically significant differences in stress levels across the three age groups.

Figure 10 provides a visual representation of this difference. The box plot suggests a trend of increasing stress levels with age, which supports the results from the ANOVA.

To further investigate the specific group differences identified by the ANOVA and to quantify the pattern observed in the box plot, we proceed with a Tukey HSD post-hoc analysis (Costa et al., 2025a).

Tukey HSD

The results of the Tukey HSD post-hoc test are presented in Table 4.

All pairwise comparisons resulted in p-values below 0.05, indicating statistically significant differences between all age groups. The younger groups appear to experience lower stress levels. For instance, Group A (under 19 years old) scores on average 3.10 points lower than Group B (ages 20–21), and 8.27 points lower than Group C (22+ years old).

The confidence intervals for each comparison do not cross zero, further reinforcing the significance of the results. These findings confirm a progressive increase in stress levels with age among university students in our sample.

**Figure 9**

Stress scores distribution

Table 1

Shapiro-Wilk test for normality applied to the transformed scores.

	Group A		Group B		Group C	
	W-statistic	p-value	W-statistic	p-value	W-statistic	p-value
Depression	0.957	0.1518	0.9655	0.0023	0.9724	0.229
Anxiety	0.9711	0.4209	0.9693	0.005	0.9902	0.929
Stress	0.953	0.0303	0.9799	0.052	0.9693	0.1629

Table 2

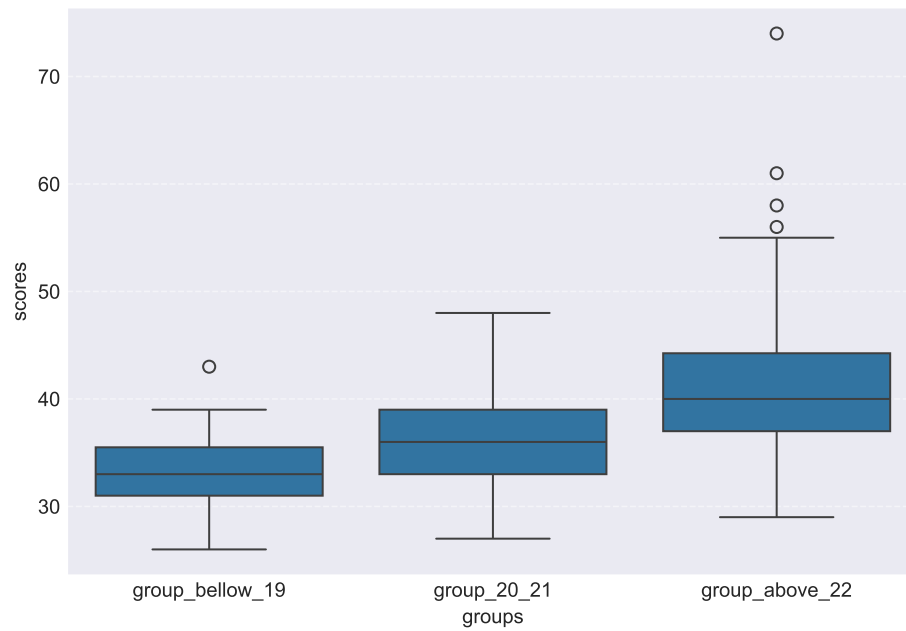
Levene test for variance applied to the scores.

	F-value	p-value
Depression	3.7803	0.0292
Anxiety	3.4976	0.0367
Stress	2.8127	0.0689

Table 3

One-Way Anova for Stress.

	sum_sq	df	F	PR(>F)
group	1728.835108	2.0	27.120712	2.963868e-11
residual	7012.052785	220.0	NaN	NaN

**Figure 10**

Stress distibution for all age groups

Table 4

Tukey's HSD Pairwise Group Comparisons (95.0% Confidence Interval).

Group 1	Group 2	meandiff	p-value	Low CI	Upper CI	Reject H ₀
Group A	Group B	-3.104	0.009	-5.563	-0.645	True
Group A	Group C	-8.270	0.000	-11.070	-5.470	True
Group B	Group A	3.104	0.009	0.645	5.563	True
Group B	Group C	-5.166	0.000	-7.298	-3.034	True
Group C	Group A	8.270	0.000	5.470	11.070	True
Group C	Group B	5.166	0.000	3.034	7.298	True

Exploratory Factor Analysis: EFA

To further investigate the underlying structure of the DASS-21 responses in our sample, we now perform an Exploratory Factor Analysis (EFA). Although the DASS-21 is designed to measure three distinct constructs, Depression, Anxiety, and Stress, we aim to empirically assess whether this three-factor structure is supported by our data, specifically within the context of Portuguese university students.

In the previous section, we conducted a one-way ANOVA to explore differences in these constructs across age groups. However, due to violations of normality and variance homogeneity in some subgroups, we were unable to draw definitive conclusions for all constructs. EFA allows us to take a step back and validate the factor structure itself, independently of group comparisons.

Our analysis is guided by the following hypotheses:

- **(H₁):** The DASS-21 items will cluster into three distinct latent factors, Depression, Anxiety, and Stress, reflecting the intended theoretical structure.
- **(H₂):** Each item will load most strongly on the factor it was originally designed to measure.

Data validation

Before conducting the Exploratory Factor Analysis (EFA), we assessed the suitability of our dataset using two standard tests: the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy and Bartlett's test of sphericity (Costa, 2025).

The results indicate excellent suitability for factor analysis. The KMO value was **0.94**, which is considered “excellent” and suggests that the variance among items is sufficiently shared to justify factor extraction. Bartlett's test of sphericity was also highly significant: $\chi^2(226) = 3096.14, p < 0.001$, confirming that the correlation matrix is not an identity matrix and that correlations between items are adequate for factor analysis.

We proceeded by examining the item correlation matrix and computing eigenvalues to determine the number of underlying factors, using the Kaiser criterion (eigenvalues > 1).

Figure 11 shows the correlation matrix for the DASS-21 items. Each item has been labeled according to its corresponding construct in the original scale (e.g., DASS_1 corresponds to Stress_1). The mapping is detailed in the replication package (Appendix A) and also on Tables 5 and 6.

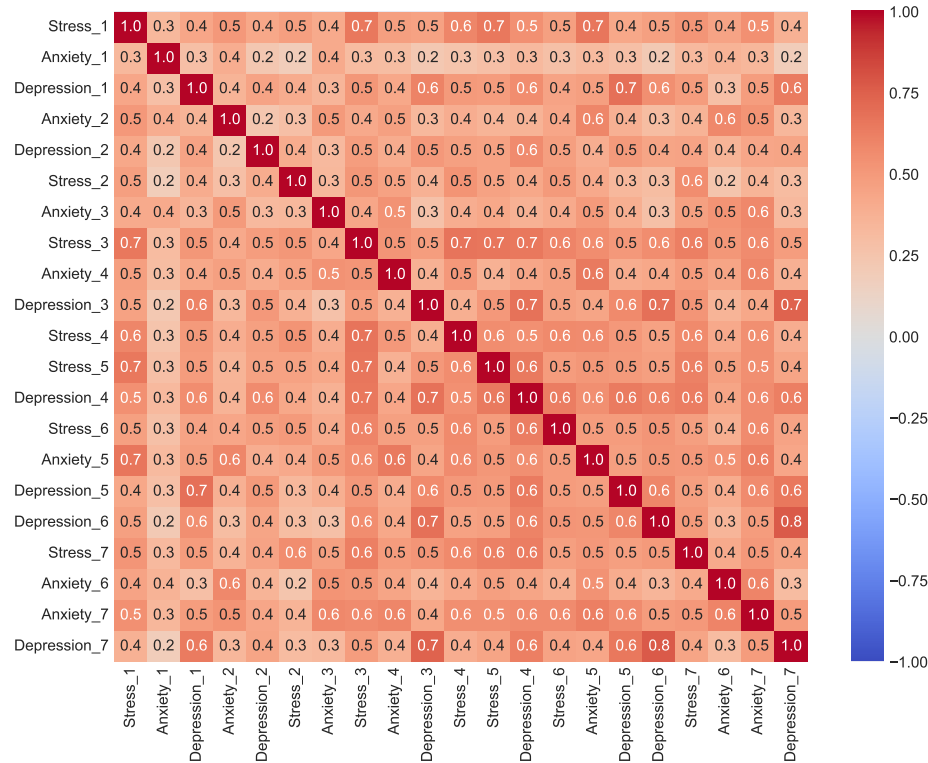


Figure 11

Correlation matrix for the DASS-21 items

The matrix reveals moderate to strong correlations ($r > 0.3$) among items within the same construct, as expected. Lower correlations (e.g., $r = 0.2$ between *Anxiety_1* and *Depression_7*) occur between items from different constructs and are not concerning given their theoretical independence. Diagonal values of 1.0 reflect perfect correlations of each item with itself.

To further assess the dimensional structure, we examined the scree plot in Figure 12.

The scree plot supports the presence of three factors. The first three components have eigenvalues of **10.383**, **1.735**, and **1.171**, respectively, each exceeding the Kaiser criterion of 1.0, while subsequent factors fall below this threshold and are therefore not retained (Santos et al.,

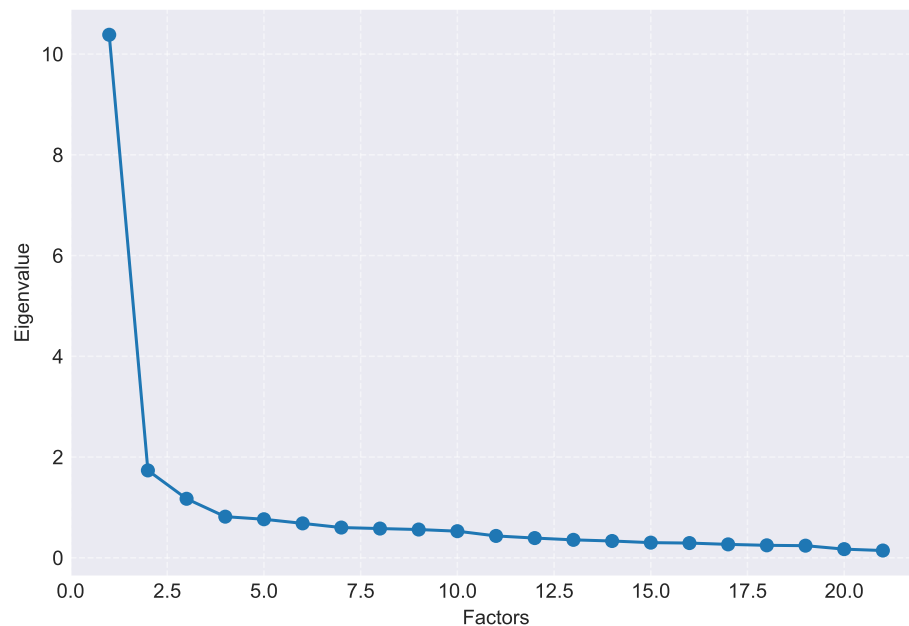


Figure 12

Scree plot for the DASS-21 items

2015).

Before proceeding with factor extraction, we visually inspected item-level distributions to detect skewness, kurtosis, and potential outliers. Figure 13 shows boxplots for all 21 items, grouped by their intended factor (Stress, Anxiety, and Depression).

Among the stress items, **Stress_1 (DASS_1)** shows minimal variability (IQR = 0), with most responses concentrated at score 2 and a few outliers. **Stress_6 (DASS_14)** displays some variability but is skewed toward lower scores, showing a different pattern from the other items on this scale. These items may be candidates for exclusion if they exhibit low communality or weak factor loadings.

For the anxiety subscale (DASS_2, DASS_4, DASS_7, DASS_9, DASS_15, DASS_19, DASS_20), distributions are generally negatively skewed, indicating that most participants reported low to moderate anxiety, with a few outliers scoring higher. These items will be evaluated for consistency and communality, especially given their limited variance.

The depression items display more balanced distributions overall. **Depression_2**

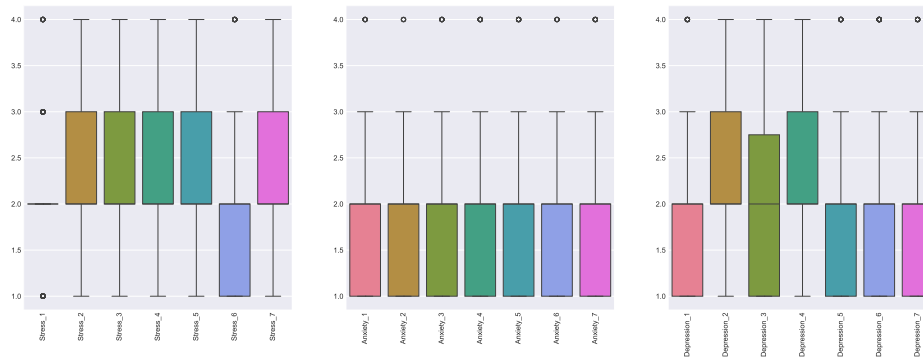


Figure 13

Data distribution breakdown by each of the factors in the DASS-21 scale.

(DASS_5) and **Depression_4 (DASS_13)** appear particularly well-distributed and may contribute strongly to the latent factor. In contrast, **Depression_3 (DASS_10)** has a broad distribution, while others (DASS_3, DASS_16, DASS_17 and DASS_21) are skewed toward lower scores. These patterns will be examined in light of factor loadings and reliability metrics, such as Cronbach's alpha (Santos et al., 2015).

This preliminary data validation confirms that the dataset is appropriate for EFA and provides an early indication of potential issues with individual items.

Analysis

Following the verification of prerequisites in the previous section and considering our sample size of 226, yielding a subject-to-item ratio greater than 10:1, we proceeded with an Exploratory Factor Analysis (EFA). This ratio meets the recommendation by Costello and Osborne (2005) for conducting reliable factor analysis.

The analysis was performed using the Python library `factor_analyzer`⁴. To explore the factor structure under different assumptions, we applied two rotation methods:

- **Varimax** (orthogonal rotation), which assumes that the extracted factors are uncorrelated;
- **Oblimin** (oblique rotation), which allows for correlation between factors.

⁴ <https://factor-analyzer.readthedocs.io/en/latest/>

Although both rotations are commonly used, oblimin is theoretically more appropriate for the DASS-21 scale, as the constructs of Depression, Anxiety, and Stress are expected to be related. Oblimin rotation typically yields more accurate communalities and reflects the overlap between psychological constructs.

The communality values resulting from both rotations are shown in Figure 14.

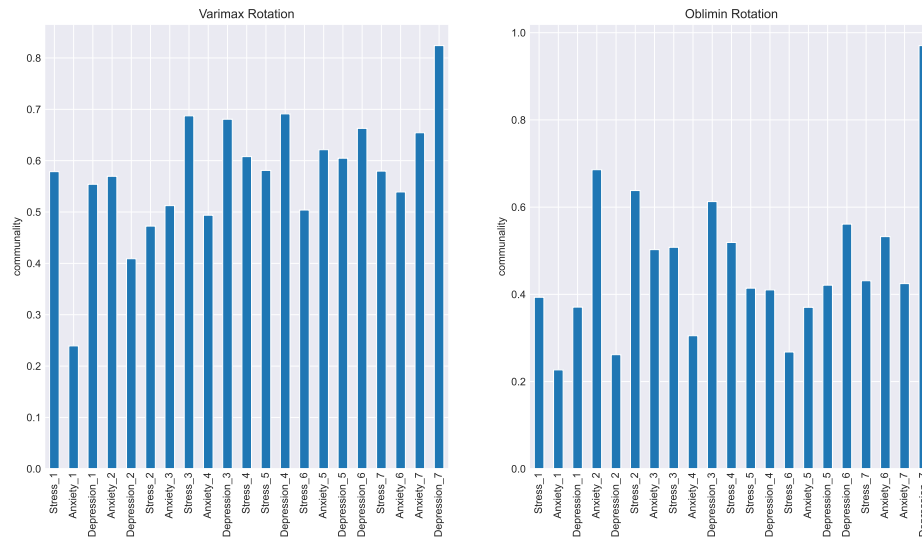


Figure 14

EFA Communalities using varimax and oblimin rotation.

Using varimax rotation, only one item, **Anxiety_1 (DASS_2)**, has a communality below the commonly accepted threshold of 0.4. With oblimin rotation, a few additional items fall below this threshold, most notably **Anxiety_1** again, which exhibits the lowest communality in both rotations. This indicates that only a small proportion of the variance in this item is explained by the extracted factors. As shown in the correlation matrix (Figure 11), this item also demonstrates weak correlation with other items, particularly those outside its intended construct, anxiety, suggesting it may not fit well within the factor structure.

Given these results, **Anxiety_1** emerges as a strong candidate for removal. However, before making this decision, we evaluated the factor loadings, presented in Tables 5 and 6, to assess whether each item contributes meaningfully to its intended factor.

Interestingly, **Anxiety_1 (DASS_2)** demonstrates an acceptable loading (>0.4) in both

rotations, despite its low communality. This suggests that while it aligns reasonably with its construct, it does not share variance well with the overall factor model. One possible explanation is ambiguity in the item's wording, though this would require further qualitative or semantic analysis.

Another notable finding involves **Depression_2 (DASS_5)**, which loads more strongly onto the Stress factor rather than the Depression factor. This pattern is also observable in the correlation matrix and could indicate that the item captures a different psychological dimension than intended. Again, further semantic review would be necessary to determine the root cause, which is outside the scope of this study.

Despite these outliers, all items demonstrate moderate to strong factor loadings and may be cautiously retained for now. A more definitive assessment would require Confirmatory Factor Analysis (CFA).

To further evaluate item reliability, we computed Cronbach's alpha for each subscale, as well as for the Anxiety subscale with and without **Anxiety_1 (DASS_2)**. Results are summarized in Table 7.

All three subscales show high internal consistency, suggesting that the items within each factor reliably measure their respective constructs. Removing **Anxiety_1** slightly increased the alpha coefficient for the Anxiety subscale, but the change (0.01) was negligible and both values fall within overlapping confidence intervals. This indicates that while **Anxiety_1** is a relatively weaker item, it does not significantly compromise scale reliability and can be retained.

The results of the Exploratory Factor Analysis provide support for both of our initial hypotheses. First, the identification of three distinct factors with eigenvalues greater than 1 and meaningful loadings across both varimax and oblimin rotations supports **H₁**, showing that the DASS-21 items cluster into three underlying constructs, Depression, Anxiety, and Stress. Second, each item loads on its intended factor, thereby supporting **H₂**. While a few items demonstrated weaker communalities or unexpected cross-loadings, particularly Anxiety_1 (DASS_2) and Depression_2 (DASS_5), these deviations were not substantial enough to undermine the overall

factor structure.

Although a Confirmatory Factor Analysis (CFA) would be the next logical step to validate the factor structure, it lies beyond the scope of this study. We conclude the factor analysis here.

Table 5*Communities and factor loadings using varimax rotation*

Item	DASS-21 item	Communality	Loading Factor 1	Loading Factor 2	Loading Factor 3
Stress 1	DASS 1	0.57	0.24	0.57	0.44
Anxiety 1	DASS 2	0.24	0.13	0.16	0.44
Depression 1	DASS 3	0.55	0.62	0.33	0.25
Anxiety 2	DASS 4	0.57	0.17	0.15	0.72
Depression 2	DASS 5	0.41	0.41	0.44	0.21
Stress 2	DASS 6	0.47	0.17	0.64	0.18
Anxiety 3	DASS 7	0.51	0.14	0.25	0.66
Stress 3	DASS 8	0.69	0.34	0.66	0.37
Anxiety 4	DASS 9	0.49	0.23	0.38	0.55
Depression 3	DASS 10	0.68	0.75	0.30	0.17
Stress 4	DASS 11	0.61	0.26	0.65	0.35
Stress 5	DASS 12	0.58	0.35	0.60	0.32
Depression 4	DASS 13	0.69	0.59	0.51	0.28
Stress 6	DASS 14	0.50	0.37	0.49	0.35
Anxiety 5	DASS 15	0.62	0.29	0.40	0.61
Depression 5	DASS 16	0.60	0.65	0.27	0.33
Depression 6	DASS 17	0.66	0.73	0.29	0.21
Stress 7	DASS 18	0.58	0.34	0.61	0.31
Anxiety 6	DASS 19	0.54	0.19	0.23	0.67
Anxiety 7	DASS 20	0.65	0.35	0.33	0.65
Depression 7	DASS 21	0.82	0.87	0.14	0.20

Table 6*Communities and factor loadings using oblimin rotation*

Item	DASS-21 item	Communality	Loading Factor 1	Loading Factor 2	Loading Factor 3
Stress 1	DASS 1	0.39	0.56	-0.02	0.27
Anxiety 1	DASS 2	0.23	0.02	0.00	0.48
Depression 1	DASS 3	0.37	0.19	0.58	0.05
Anxiety 2	DASS 4	0.69	-0.11	0.01	0.82
Depression 2	DASS 5	0.26	0.42	0.29	-0.01
Stress 2	DASS 6	0.64	0.79	-0.08	-0.09
Anxiety 3	DASS 7	0.50	0.06	-0.06	0.70
Stress 3	DASS 8	0.51	0.70	0.08	0.11
Anxiety 4	DASS 9	0.31	0.26	0.02	0.49
Depression 3	DASS 10	0.61	0.13	0.77	-0.06
Stress 4	DASS 11	0.52	0.71	-0.01	0.10
Stress 5	DASS 12	0.41	0.63	0.12	0.07
Depression 4	DASS 13	0.41	0.44	0.46	0.02
Stress 6	DASS 14	0.27	0.47	0.19	0.16
Anxiety 5	DASS 15	0.37	0.25	0.07	0.55
Depression 5	DASS 16	0.42	0.07	0.62	0.18
Depression 6	DASS 17	0.56	0.10	0.74	0.00
Stress 7	DASS 18	0.43	0.64	0.11	0.06
Anxiety 6	DASS 19	0.53	0.01	0.00	0.73
Anxiety 7	DASS 20	0.42	0.11	0.17	0.62
Depression 7	DASS 21	0.97	-0.14	0.98	0.03

Table 7*Cronbach's α and confidence intervals for each factor*

Factor	Cronbach's α	Confidence interval
Stress	0.90	[0.877 0.917]
Depression	0.91	[0.892 0.928]
Anxiety	0.87	[0.846 0.897]
Anxiety (w/ DASS_2)	0.88	[0.853 0.902]

Cluster analysis

As established in the Exploratory factor analysis, our dataset accurately reflects the intended structure of the DASS-21 scale, capturing the three psychological constructs of depression, anxiety, and stress. With the One-Way Anova analysis, we attempted to identify psychological profiles based on age groups. However, the results, particularly for depression and anxiety, were inconclusive, as we failed to reject the null hypotheses, indicating no statistically significant differences across age categories.

To gain a deeper and potentially more informative understanding of the psychological diversity within our sample, we now turn to clustering analysis (Barbosa, 2025). This exploratory approach allows us to group participants based on the similarity of their psychological profiles, regardless of predefined demographic categories such as age. For this purpose, we use the factor scores derived from the Exploratory Factor Analysis, representing each participant's position along the depression, anxiety, and stress dimensions.

We begin this analysis with the following hypothesis:

- **H₁:** There are distinct psychological profiles among university students based on their depression, anxiety, and stress patterns.

The analysis proceeds in two stages: first, we perform a hierarchical clustering to explore the underlying structure and determine a suitable number of clusters; second, we apply a non-hierarchical method (k-means clustering) to assign individuals to specific psychological profile groups based on the factor scores.

Analysis

We began this analysis by calculating individual factor scores for stress, depression, and anxiety, followed by standardizing the values. These standardized scores were used as input for clustering, and the detailed computations can be found in the replication package accompanying this work.

Next, we applied Ward's hierarchical clustering method, which minimizes within-cluster

variance and typically produces compact, similarly sized clusters⁵. This approach allows for the construction of a dendrogram, shown on Figure 15, which visually represents the distances between merged clusters. For easier interpretation, the dendrogram is truncated to display only the last 50 merges.

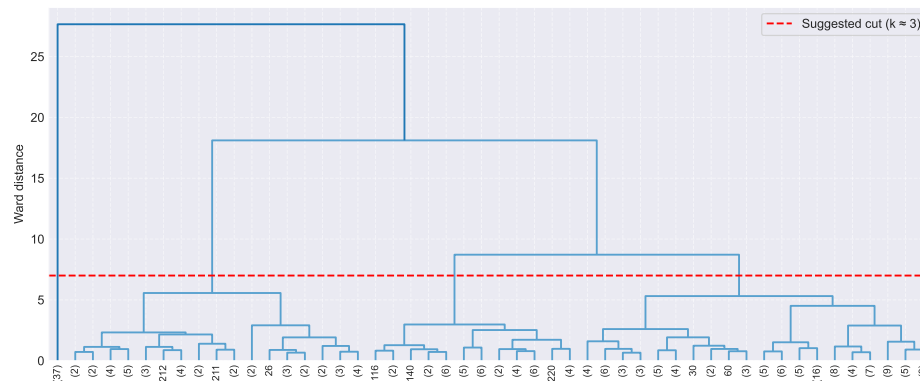


Figure 15

Hierarchical clustering dendrogram, truncated to the last 50 merges.

By examining the largest vertical distance between merges, highlighted by the red dashed line, we can estimate that a three-cluster solution is appropriate. This initial assumption is further supported by the elbow and silhouette plots shown in Figure 16.

Both methods indicate that three is an optimal number of clusters. The elbow plot shows a clear bend at $k = 3$, and the silhouette score reaches its maximum at the same value. Based on this, we proceeded with a k-means clustering algorithm using $k = 3$.

To visualize the resulting clusters, we applied Principal Component Analysis (PCA) (Frades & Matthiesen, 2009) to reduce the dimensionality of the data. The projected clusters are shown in Figure 17.

As can be observed, the clusters are well separated, indicating that the psychological profiles captured by the DASS-21 scores form meaningful groupings. However, Cluster 2 appears as a singleton, likely representing an outlier or an overfitted group. To explore this further, we also ran k-means with $k = 2$, as shown in Figure 18. While this removed the singleton, one data

⁵ According to the documentation in <https://scikit-learn.org/stable/modules/clustering.html#hierarchical-clustering>

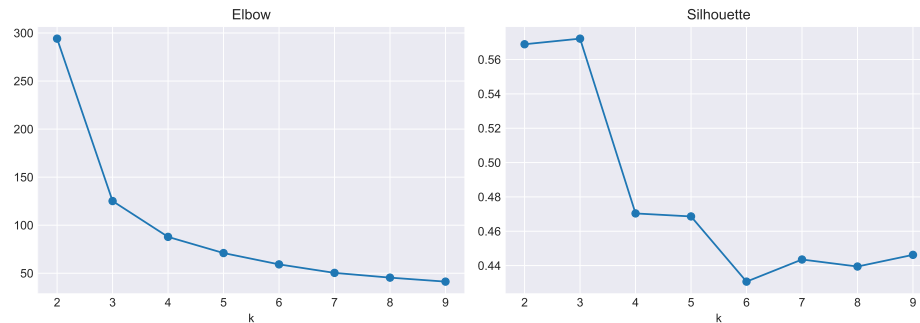


Figure 16

Elbow and Silhouette plots for the hierarchical clustering analysis.

point still remains isolated from the rest of its cluster, reinforcing the likelihood of an outlier or maybe a rare psychological or mental profile. For simplicity we will treat as an outlier.

For the remainder of the analysis, we focus on Clusters 0 and 1, which represent more cohesive and substantial groups. To better understand the psychological makeup of these clusters, we profiled them based on the mean and standard deviation of the factor scores. Cluster 2 (the singleton) was excluded to avoid distortion and noise in the analysis. The resulting boxplots are presented in Figure 19.

As expected, two distinct psychological profiles emerge. Cluster 0 shows low median scores across all three scales—indicating a generally low level of psychological distress, while Cluster 1 has higher medians and greater variability, suggesting elevated risk for mental health concerns. These findings support our initial hypothesis:

- **H₁:** There are distinct psychological profiles among university students based on their depression, anxiety, and stress patterns.

Although further statistical testing (e.g., T-tests) could offer additional insights into the significance of differences between the groups, such analysis lies beyond the scope of this work. We will therefore conclude our clustering analysis here.

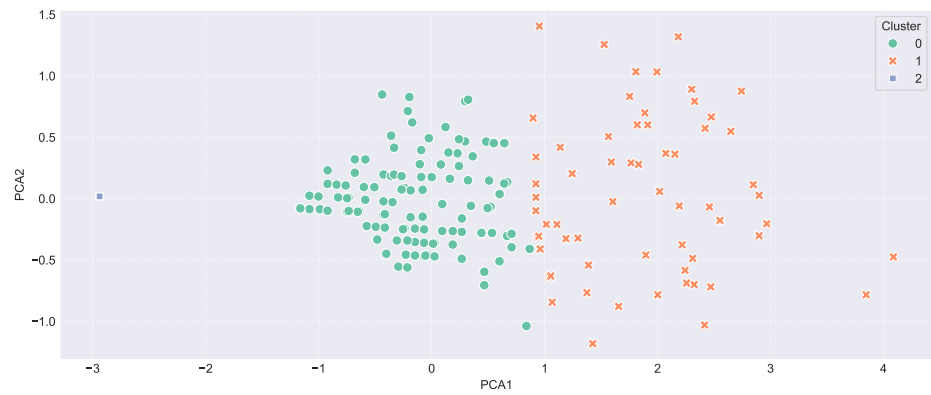


Figure 17
Clusters visualization after PCA reduction.



Figure 18
Clusters visualization after PCA reduction using only 2 clusters.

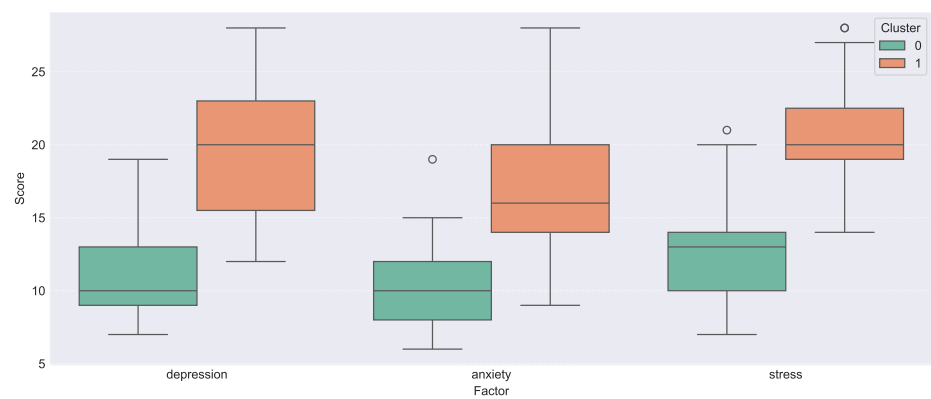


Figure 19
Clusters profile score.

Conclusion

In this study, we conducted a comprehensive analysis of psychological distress among Portuguese university students using the DASS-21 scale. Our goal was to better understand how depression, anxiety, and stress manifest in this population, through three complementary statistical approaches: inferential testing (ANOVA), dimensionality assessment (Exploratory Factor Analysis), and unsupervised learning (clustering).

First, we used **one-way ANOVA** to investigate whether levels of depression, anxiety, and stress varied significantly across different age groups. While the analysis revealed a statistically significant difference in stress levels, particularly between students aged 18–20 and those over 24, it did not show meaningful variation in depression or anxiety across age groups. These results suggest that age may influence stress perception more than other types of psychological distress in this sample, but that broader age-related patterns for depression and anxiety may be more complex or influenced by other variables.

Second, we conducted an **Exploratory Factor Analysis (EFA)** to validate the structure of the DASS-21 scale. Using both orthogonal (varimax) and oblique (oblimin) rotations, we confirmed a three-factor structure consistent with the original theoretical model. Although some items exhibited low communalities—particularly *DASS_2*, the overall factor loadings were acceptable, and internal consistency remained high across all subscales. Cronbach's alpha values supported the reliability of the three constructs, allowing us to retain all items for further analysis. These results reinforce the validity of the DASS-21 structure in the Portuguese university student population and support its use for assessing psychological distress.

Finally, we applied clustering techniques to identify latent psychological profiles among students. A combination of hierarchical clustering, k-means, and dimensionality reduction through PCA revealed the presence of two primary psychological profiles: one group exhibiting lower scores across all distress dimensions, and another showing elevated levels of depression, anxiety, and stress. This result confirmed our hypothesis that meaningful psychological groupings exist within the student population. The clustering analysis also highlighted a potential outlier

with a potential unusual distress profile.

In summary, this study provides evidence that:

- Stress varies significantly with age in university students, while depression and anxiety do not.
- The DASS-21 scale reliably measures three distinct but related psychological constructs in this population.
- There are clearly identifiable psychological profiles among students, ranging from low to high distress, which may have implications for targeted mental health interventions.

Future work could expand on this by incorporating confirmatory factor analysis and other statistical analysis to better profile the individuals and define tailored interventions based on psychological profiles.

Appendix A

Replication Package

This appendix describes the structure and usage of the replication package created for this study. All Jupyter notebooks, datasets, and generated figures are available at:

https://github.com/ricardojaferreira/dass_21_replication_package

The repository is organized as follows:

- **/data_analysis:** Contains the Jupyter notebooks used to conduct the statistical analyses presented in this work:
 - *demographic_overview.ipynb* – Generates the demographic profile discussed in the introduction.
 - *anova.ipynb* – Performs the One-Way ANOVA analysis.
 - *exploratory_factor_analysis.ipynb* – Runs the exploratory factor analysis.
 - *cluster_analysis.ipynb* – Implements the clustering analysis.

This folder also includes the dataset used throughout the analyses: *BD_MI_II_2024_25.sav*.

- **/figures:** Contains the figures generated by the analysis scripts. Each subfolder corresponds to a specific analysis section: *demographic_analysis*, *anova*, *efa*, and *cluster*.
- **Root directory:** Includes a PDF version of this document for reference.

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